

High performance Analog and Mixed signal VLSI System design

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Switched capacitor circuits

Switched capacitor circuits

- CMOS process technologies have very good transistor devices - especially if the transistors are only used as switches.
- The process technology allows construction of good quality capacitors as well using the metal layers in the process.
- It is however quite difficult to get large resistors and inductors with good quality factor in these technologies.
- Even for circuits which require small values of resistors setting time constants accurately has been challenging due to the large process variations in resistors. Process variations in resistors are not in sync with the variations in other components.
- Switched capacitor circuits are one approach to design of analog circuits like filters, amplifiers etc. which do not suffer from these limitations.

The switched capacitor resistor

- A resistor controls the current flowing between two terminals based on its resistance value. In other words it determines the rate of flow of charge between two terminals of different potentials.

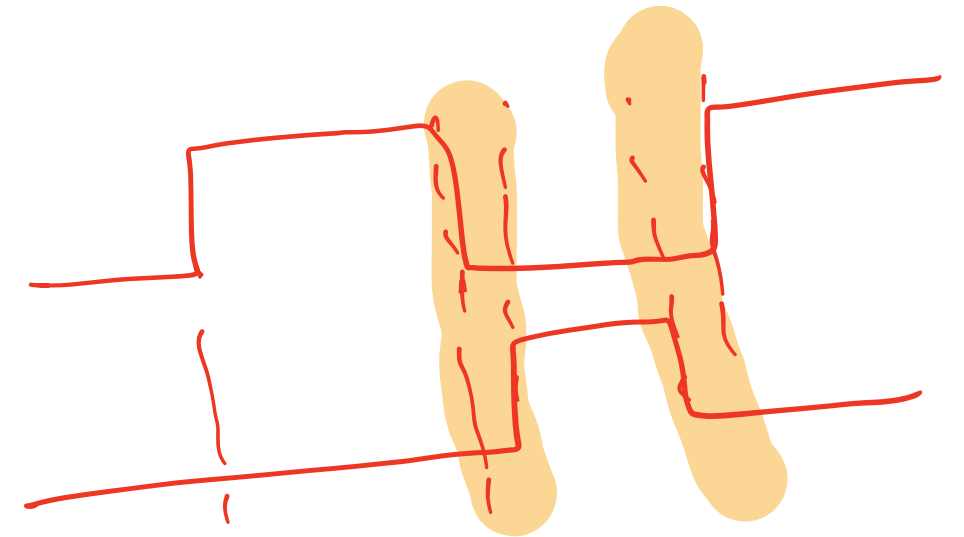
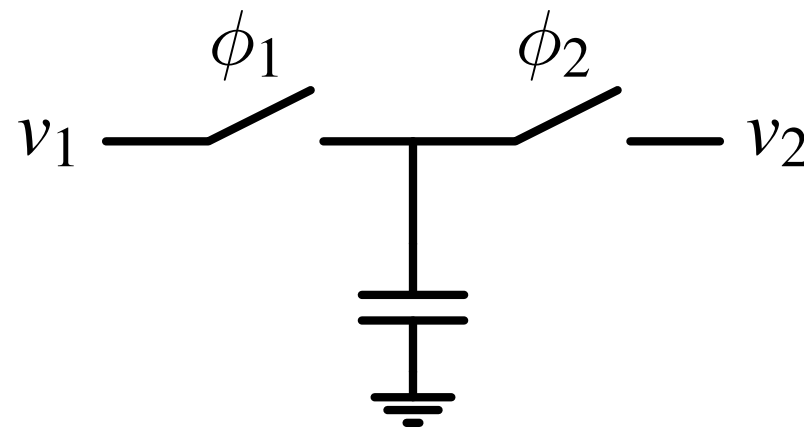
$$I = \frac{dQ}{dt} = \frac{V_1 - V_2}{R}$$

The switched capacitor resistor

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$$I = \frac{dQ}{dt} = \frac{V_1 - V_2}{R}$$

- The rate of flow of charge between two terminals can also be controlled using switches and capacitors as shown.



- ϕ_1 and ϕ_2 are non overlapping clocks.

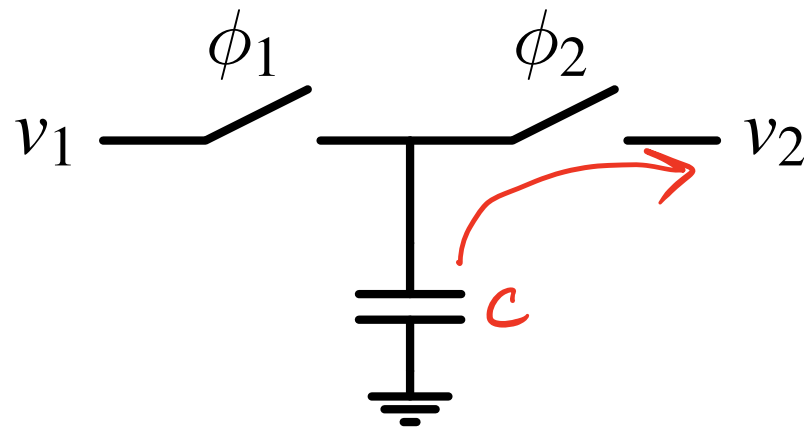
The switched capacitor resistor

- When ϕ_1 is high, the capacitor charges to v_1 and the charge on the capacitor is

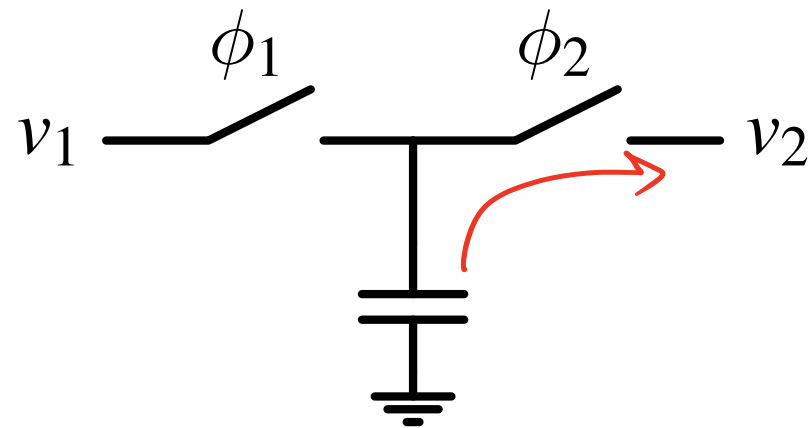
$$Q_1 = Cv_1$$

- When ϕ_2 is high, the capacitor charges to v_2 and the charge on the capacitor is

$$Q_2 = Cv_2$$



The switched capacitor resistor



- When ϕ_1 is high, the capacitor charges to v_1 and the charge on the capacitor is

$$Q_1 = Cv_1$$

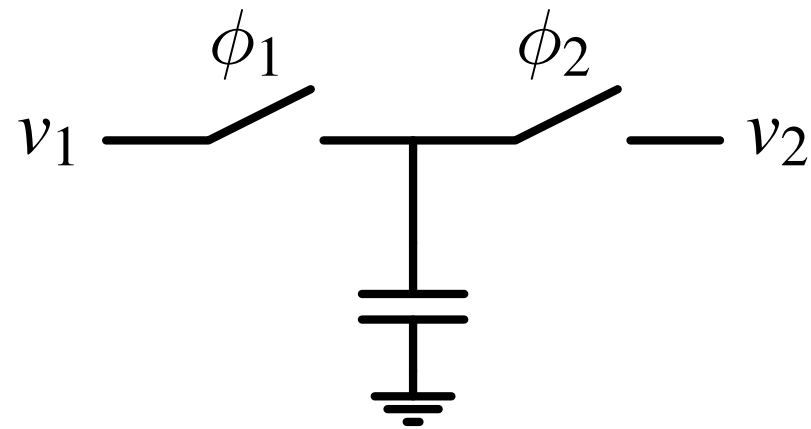
- When ϕ_2 is high, the capacitor charges to v_2 and the charge on the capacitor is

$$Q_2 = Cv_2$$

- The change in charge after this sequence is the charge drawn from v_2 , which is

$$\Delta Q = C(v_2 - v_1)$$

The switched capacitor resistor



- This ΔQ is drawn from v_2 in every clock cycle. Therefore the rate of flow of charge can be written as

$$I = \frac{dQ}{dt} = \frac{C(v_2 - v_1)}{T} = fC(v_2 - v_1)$$

$f = \frac{1}{T}$

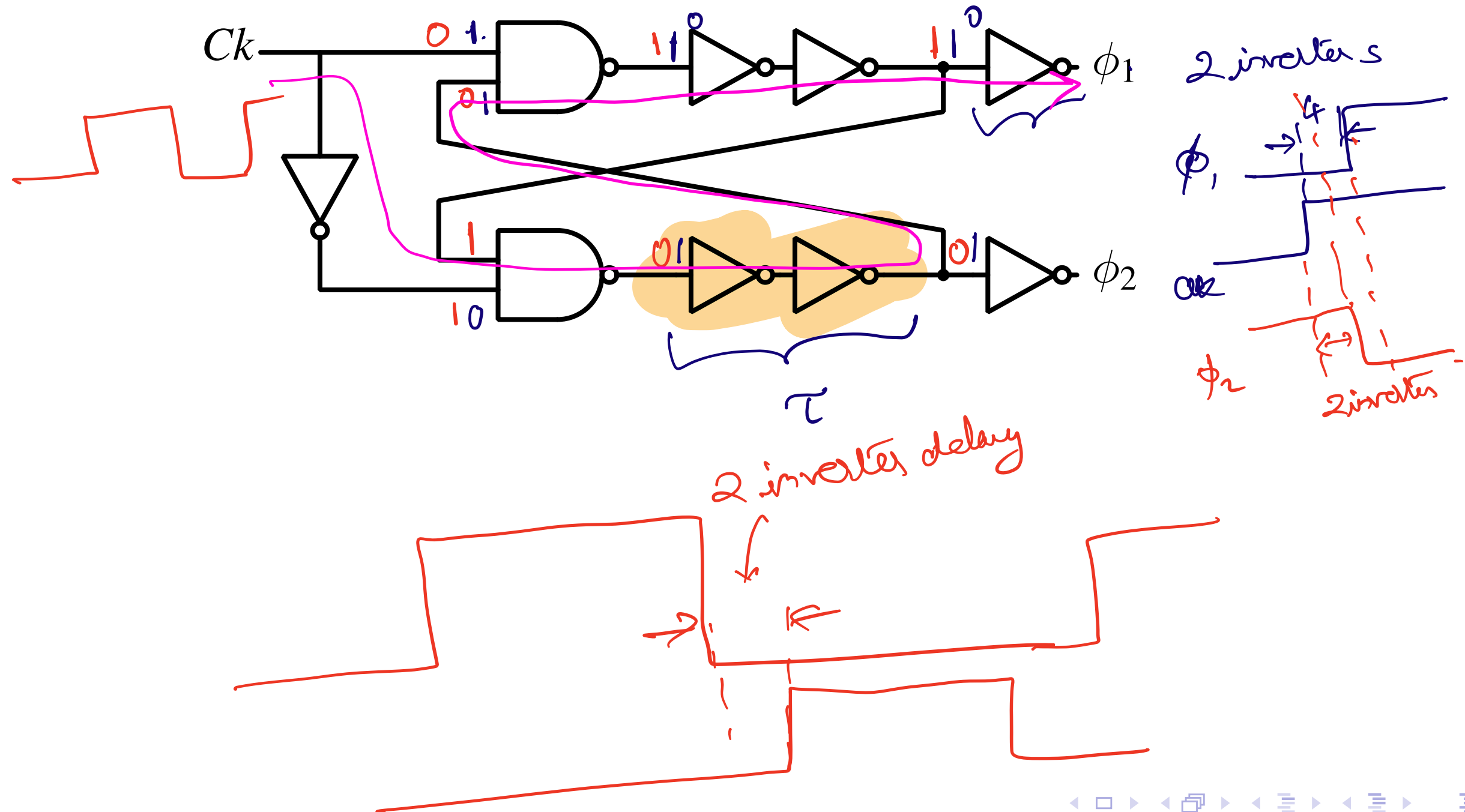
- Hence, the effective resistance is

$$R = \frac{1}{fC}$$

- If the switching frequency is much larger than the signal frequency, the approximation of this circuit as a resistance is good.

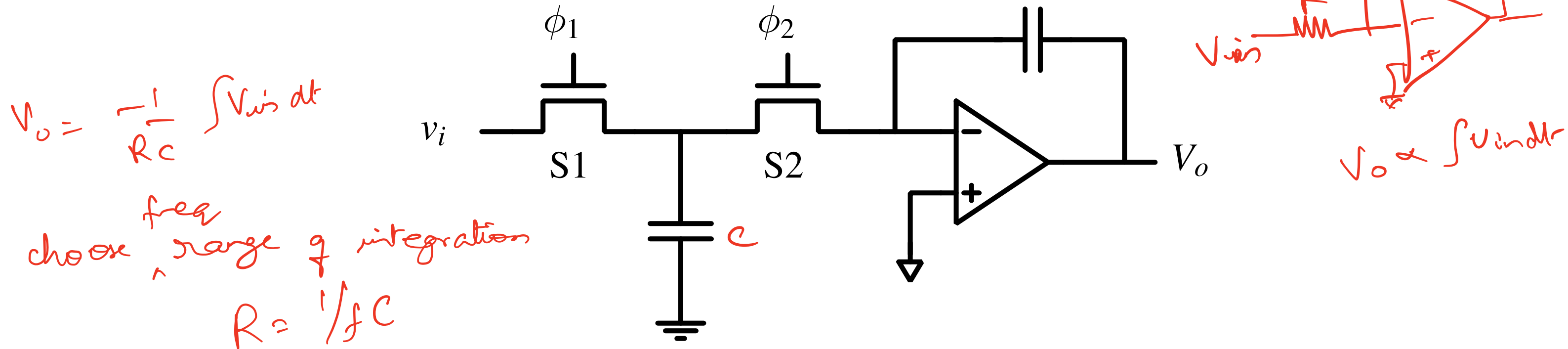
How to generate the non-overlapping clocks

- Non overlapping clocks can be generated using a simple digital circuit as shown.



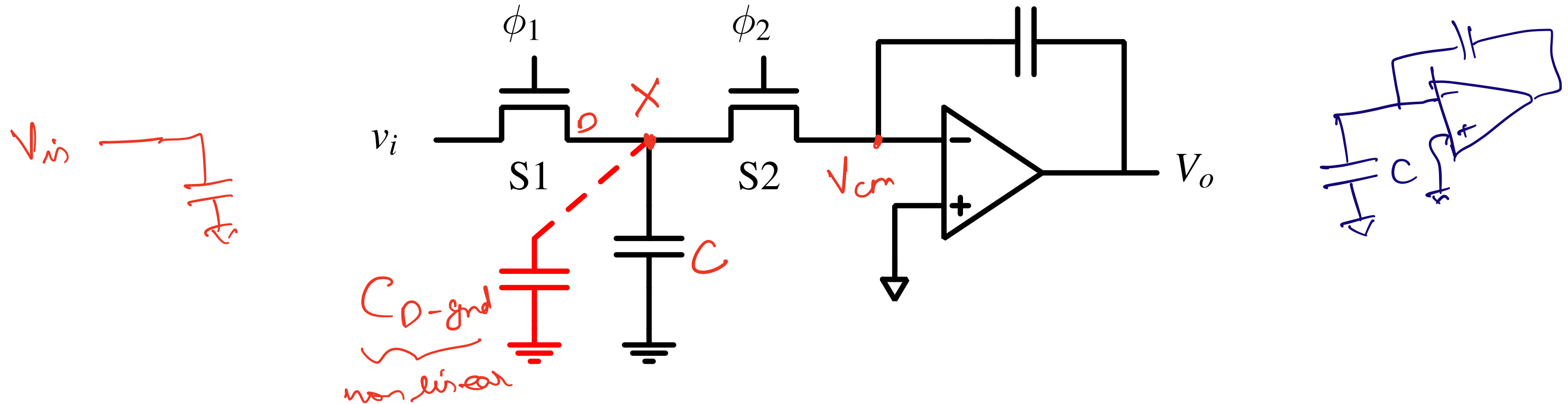
Integrator circuit

- Consider a simple integrator constructed using an opamp, in which the resistor is replaced with a switched capacitor resistor.



- This basic circuit will work as an integrator. However, it will have problems of charge injection and parasitic capacitances of different nodes.

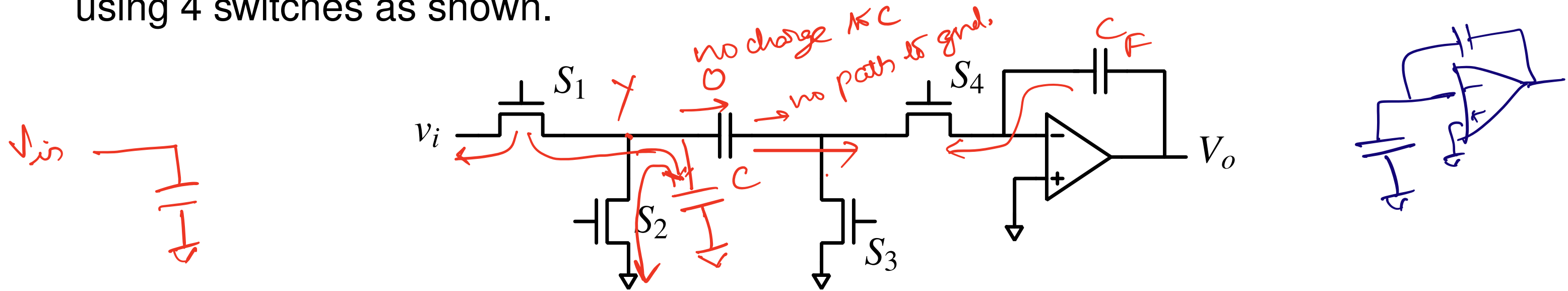
Integrator circuit



- The charge injection by S_1 depends on the input and introduces non-linearity in the output.
- The non-linear parasitic capacitances of the switches results in non-linearity in the output.

Parasitic insensitive integrator

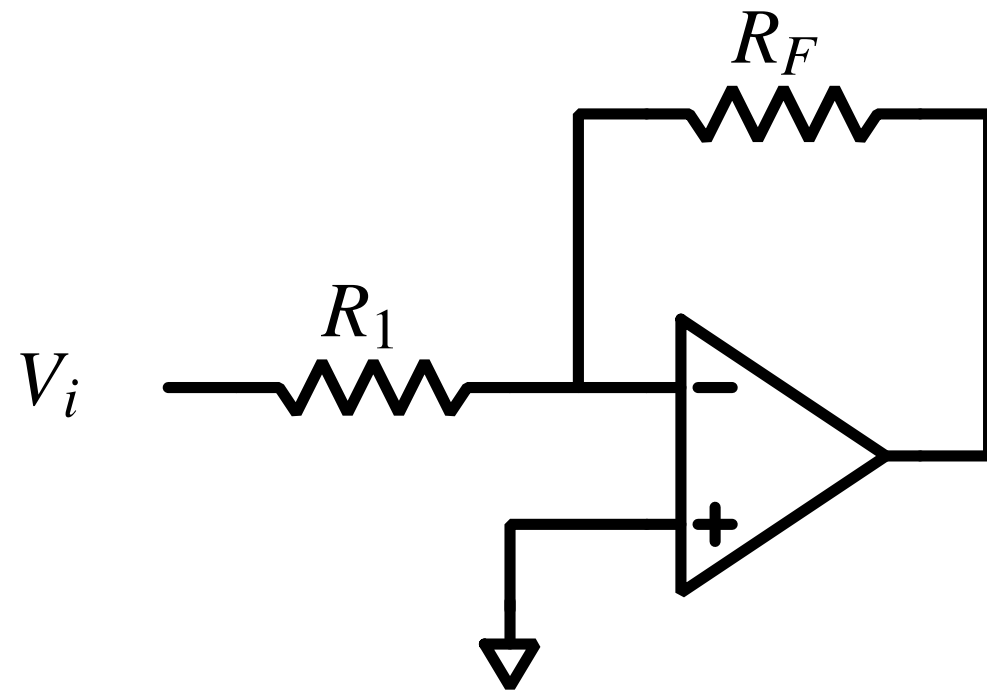
- The problems of parasitic capacitance and charge injection can be addressed by using 4 switches as shown.



- During sampling phase, S_1 and S_3 are ON. S_2 and S_4 are OFF.
- First S_3 turns off. The charge injection is of a fixed amount. *DS of S_3 are at gnd.*
- S_1 turns off next. The charge injected by S_1 cannot go to C_1 as there is no path to ground. This charge goes to the parasitic capacitance on the node.
- S_2 and S_4 turn on next to complete the cycle.

Example: Implementing feedback in an amplifier

- Consider a simple opamp based inverting amplifier as shown.

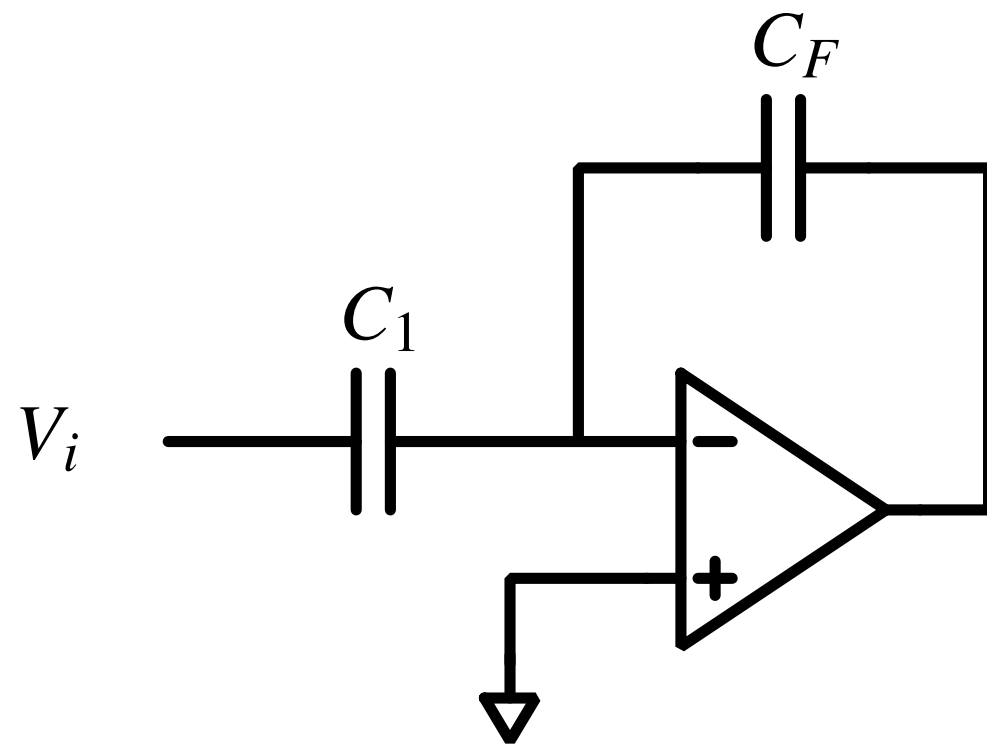


$$= -\frac{R_F}{R_1}$$
$$= -\frac{Z_F}{Z_1}$$

- Is it possible to design this amplifier without using resistors?

Example: Implementing feedback in an amplifier

- Consider a simple opamp based inverting amplifier as shown.

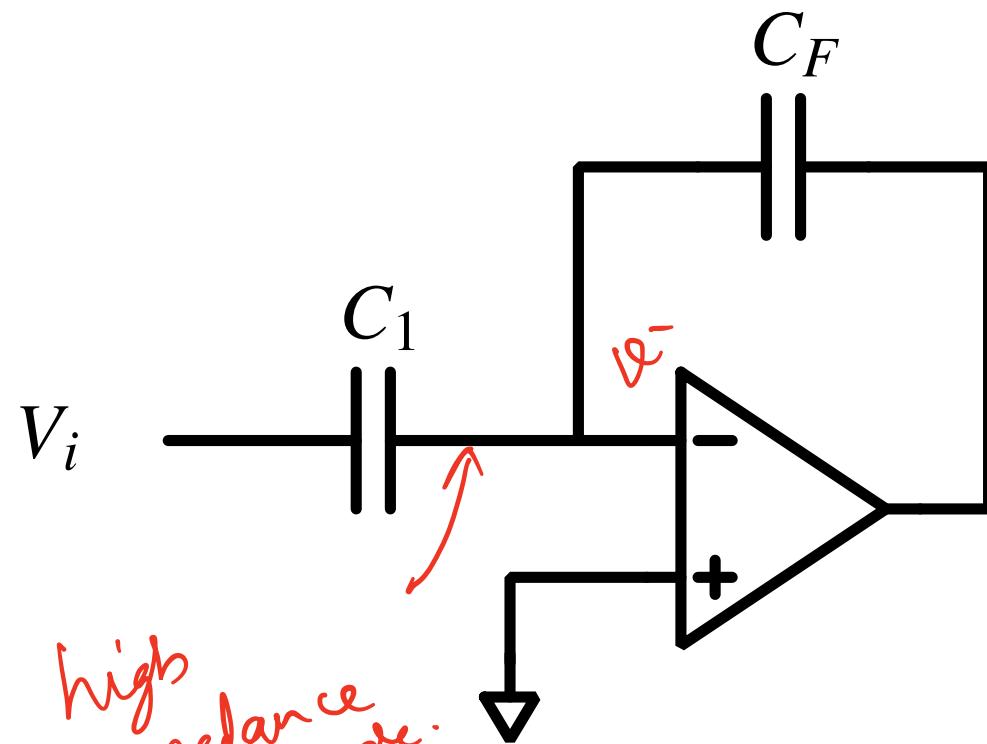


$$= \frac{-Z(C_F)}{Z(C_1)} \approx -\frac{C_1}{C_F}$$

- Is it possible to design this amplifier without using resistors?

Example: Implementing feedback in an amplifier

- Consider a simple opamp based inverting amplifier as shown.



operating pt is not defined.

- Is it possible to design this amplifier without using resistors?
- This will not work as the operating point for the opamp is not set.

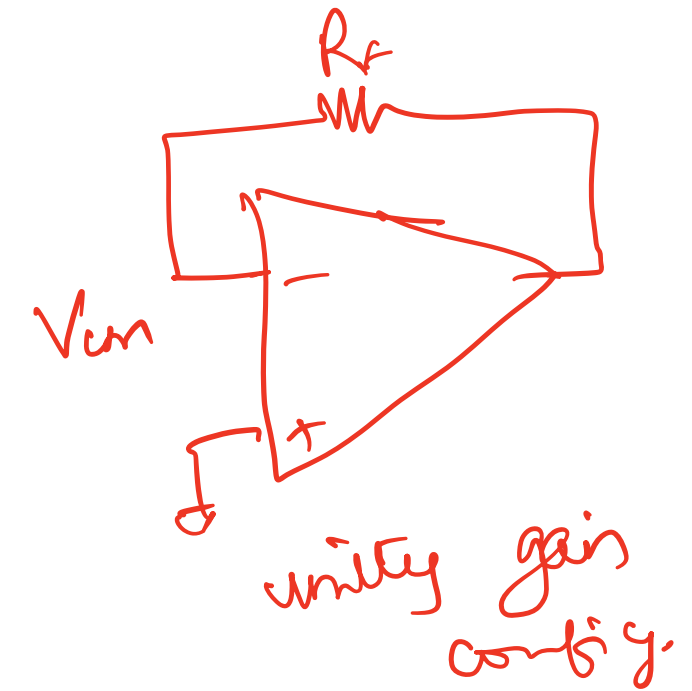
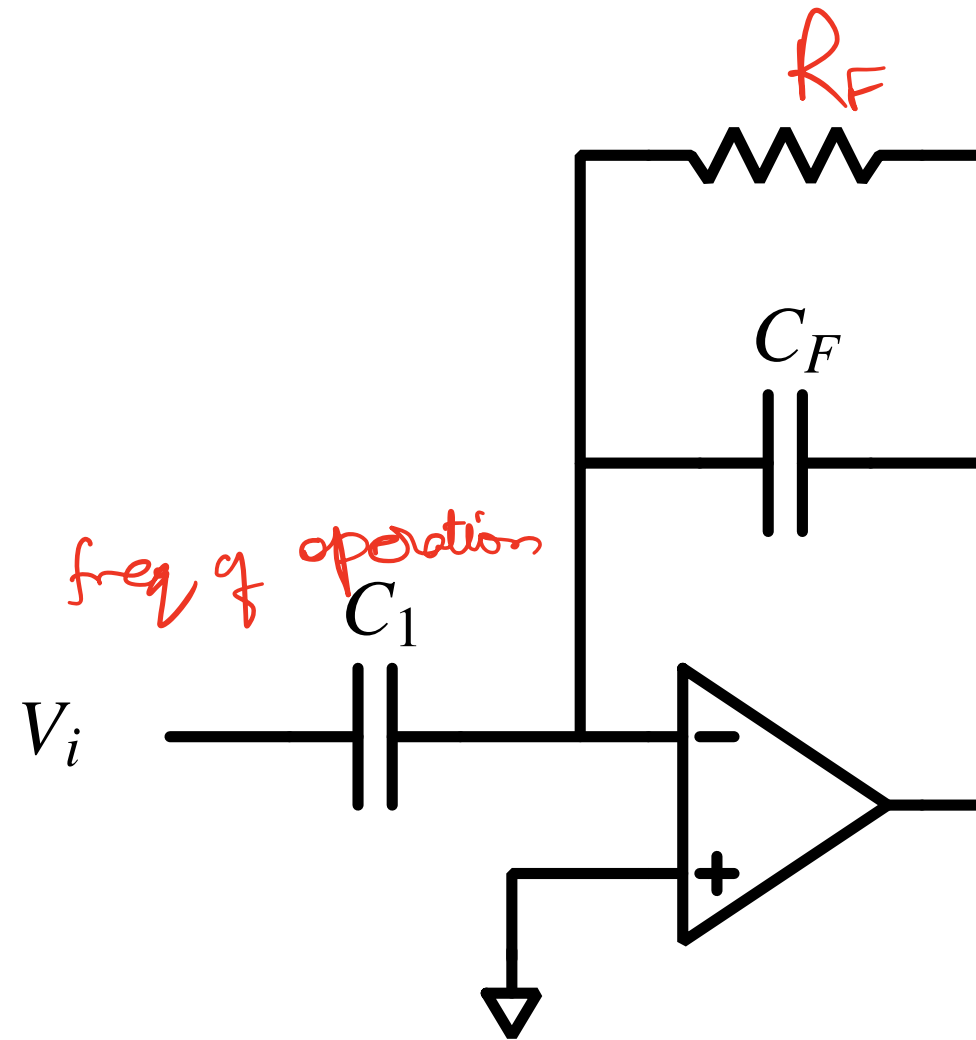
Opamp feedback using capacitive networks

- We can set the operating point by using a feedback resistor, in addition to the capacitor.

$$\text{gain} = \frac{R_F \parallel \frac{1}{sC_F}}{\frac{1}{sC_1}}$$

if $Z(R_F) \Rightarrow Z(C_F)$ @ freq of operation

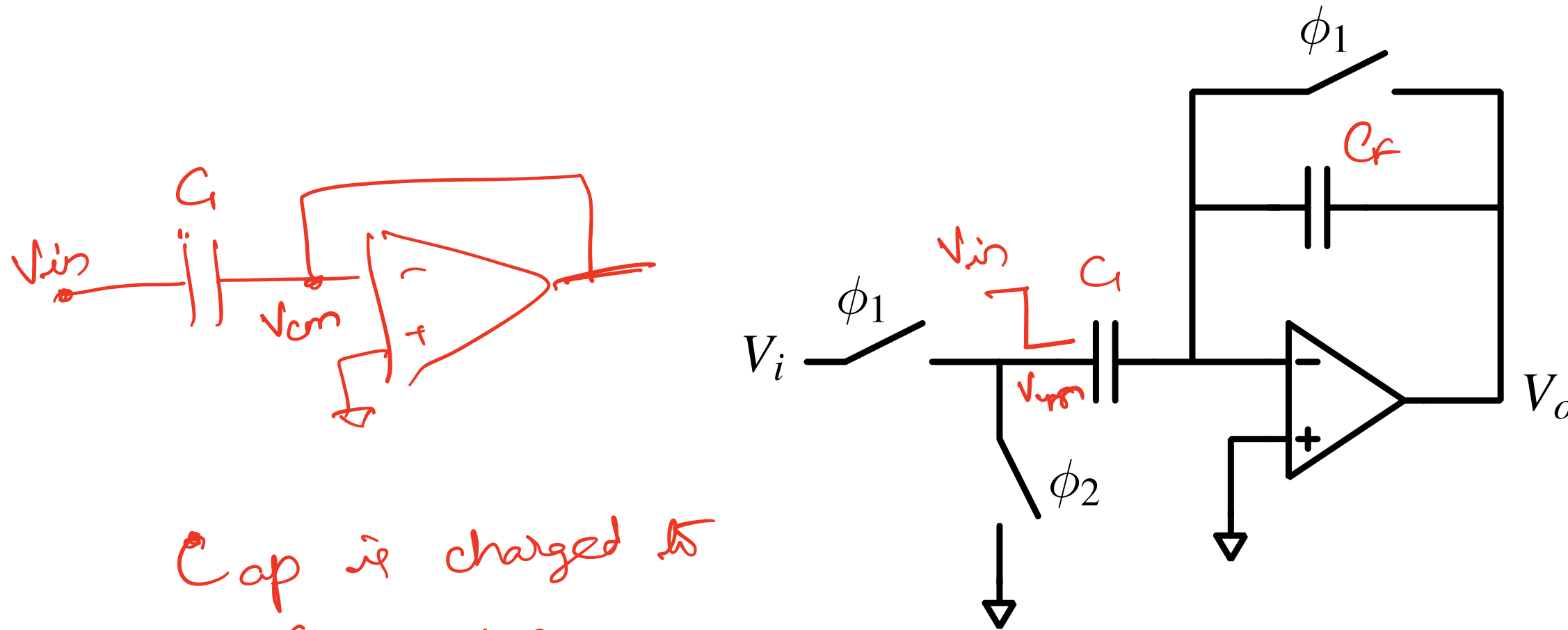
$$\text{gain} \approx \frac{C_1}{C_F}$$



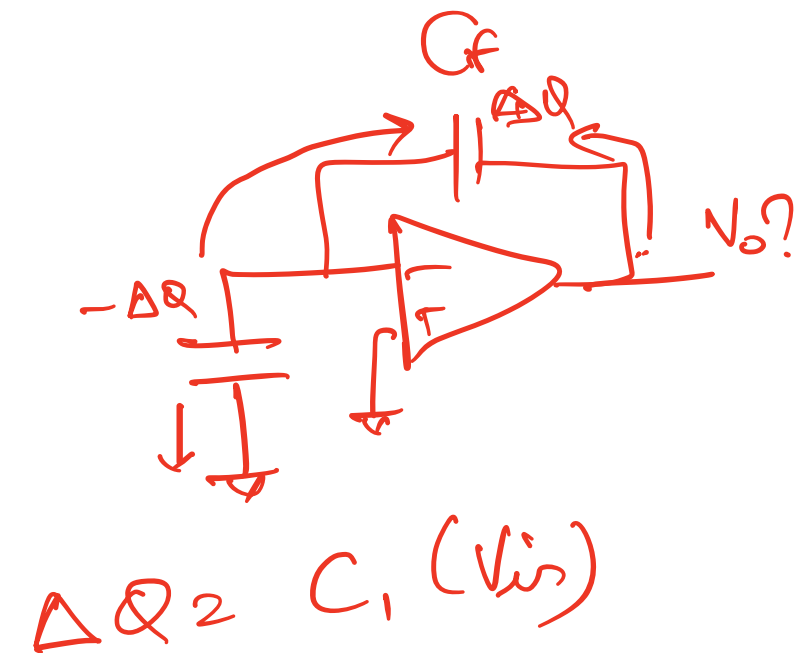
- The performance of the circuit is relatively insensitive to the value of the resistor.
- However, this amplifier works only for AC signals.

Opamp feedback using capacitive networks

- We can design an amplifier feedback network using only capacitors and switches as



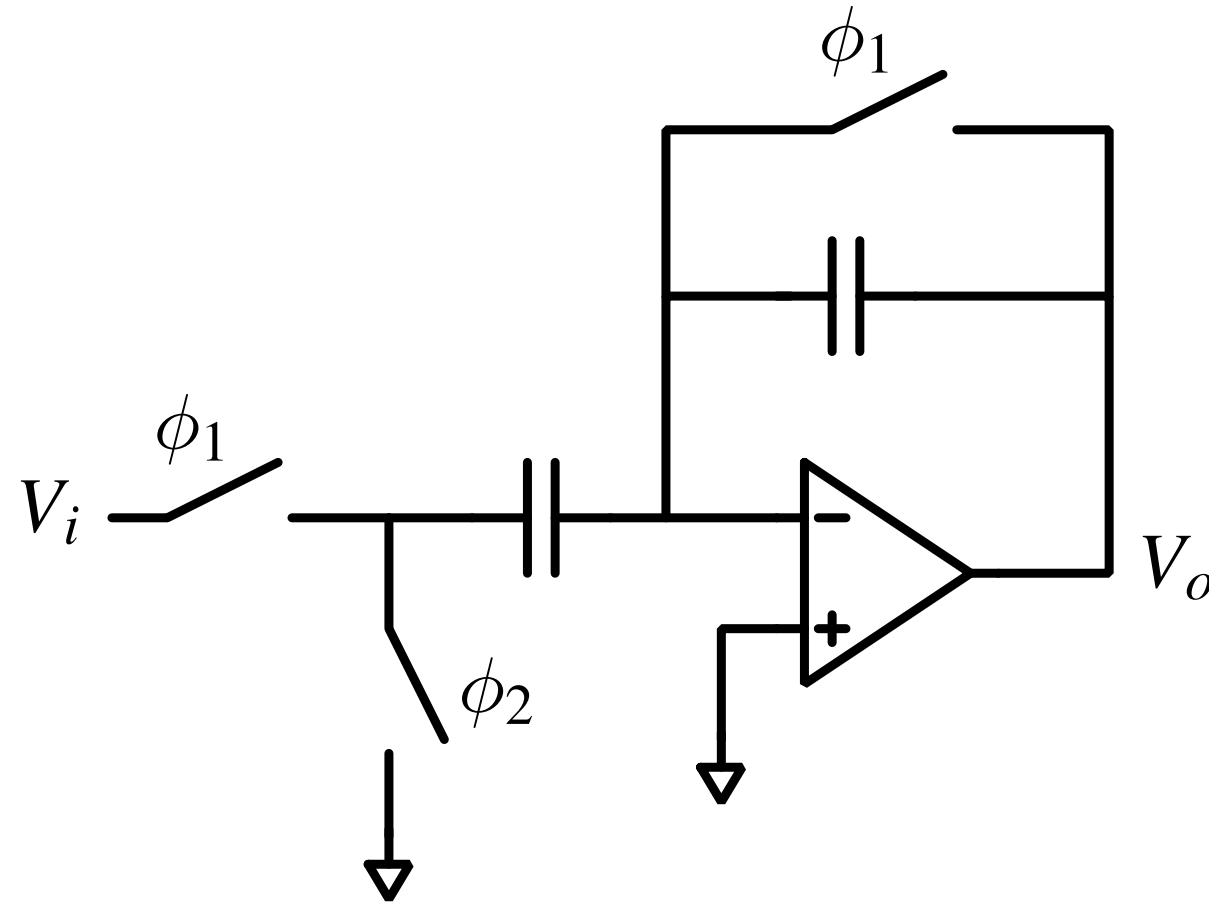
Cap is charged to
 $(V_{in} - V_{cm})$
 $Q_1 = C_1 (V_{in} - V_{cm})$



$$V_o = \frac{C_1}{C_f} V_{in}$$

Opamp feedback using capacitive networks

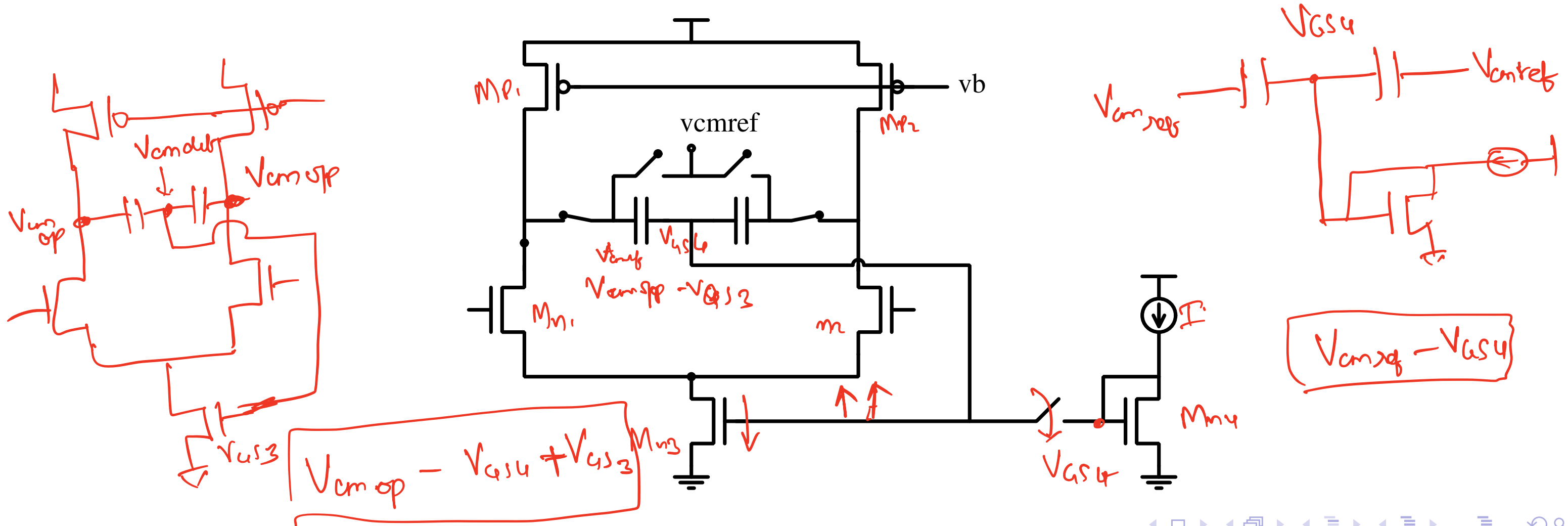
- We can design an amplifier feedback network using only capacitors and switches as



- Notice that this circuit works for DC signals as well.

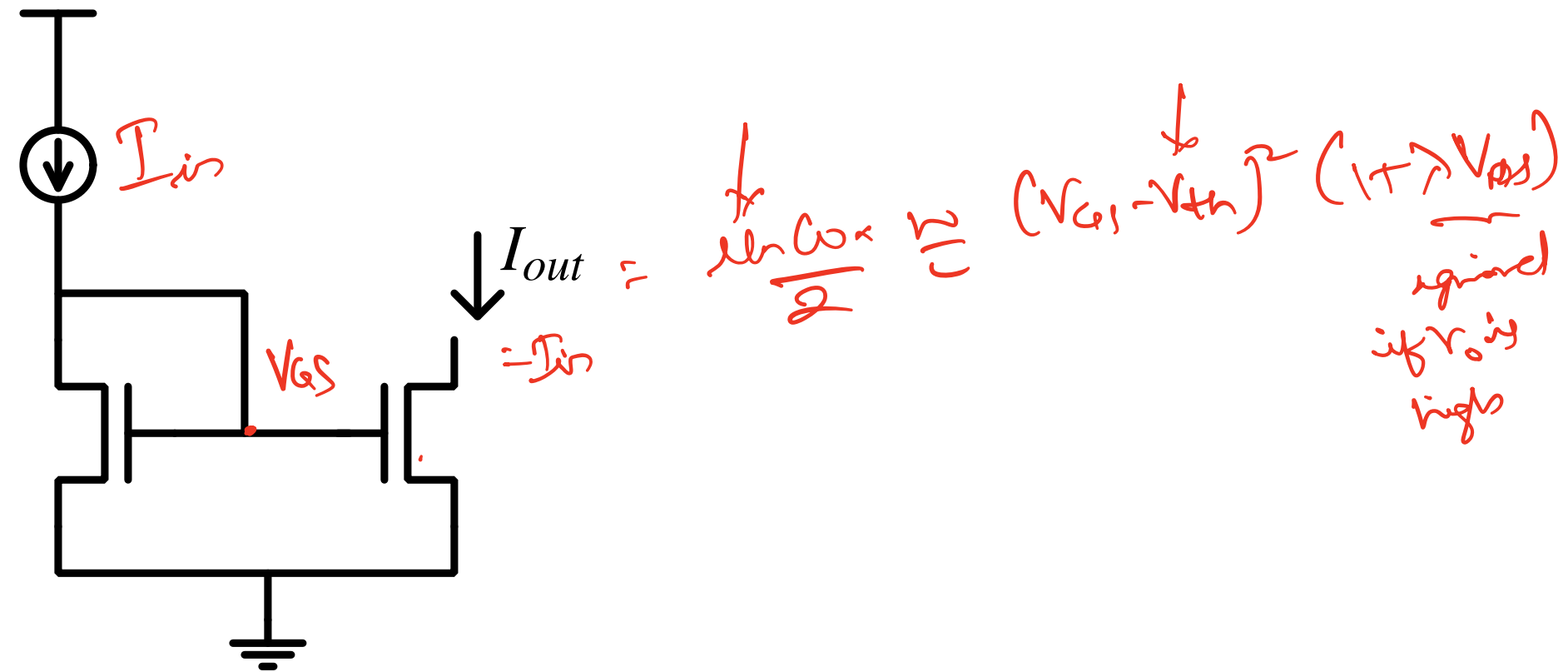
Switched capacitor CMFB

- Recall our discussions on Fully differential amplifiers wherein we needed a CMFB circuit with high dynamic range.
- We can construct CMFB circuits using switched capacitors which give high dynamic range if we have good switches.



Switched capacitor circuits can be used to improve current mirror matching

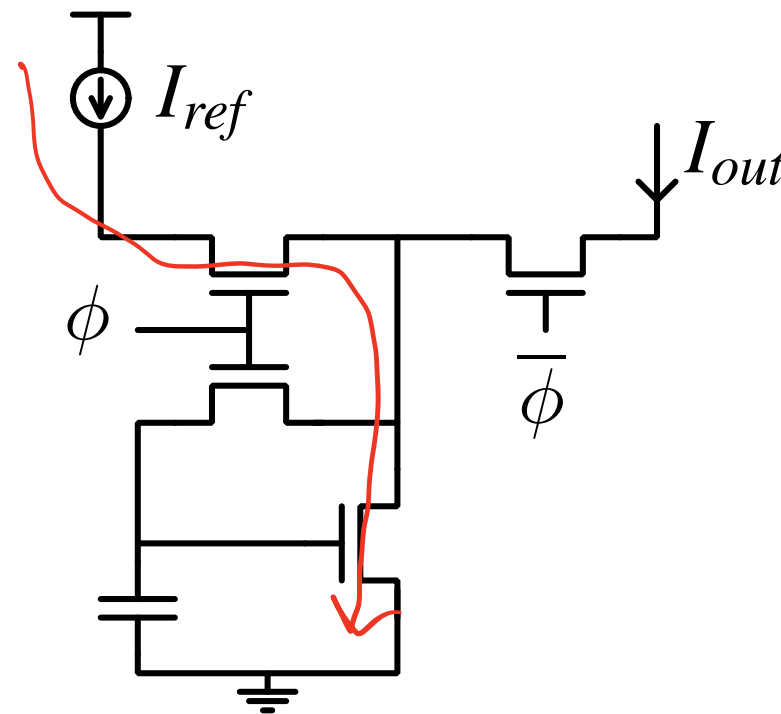
- Consider a simple current mirror as shown.



- The accuracy of the mirror depends on the matching of the two transistors.

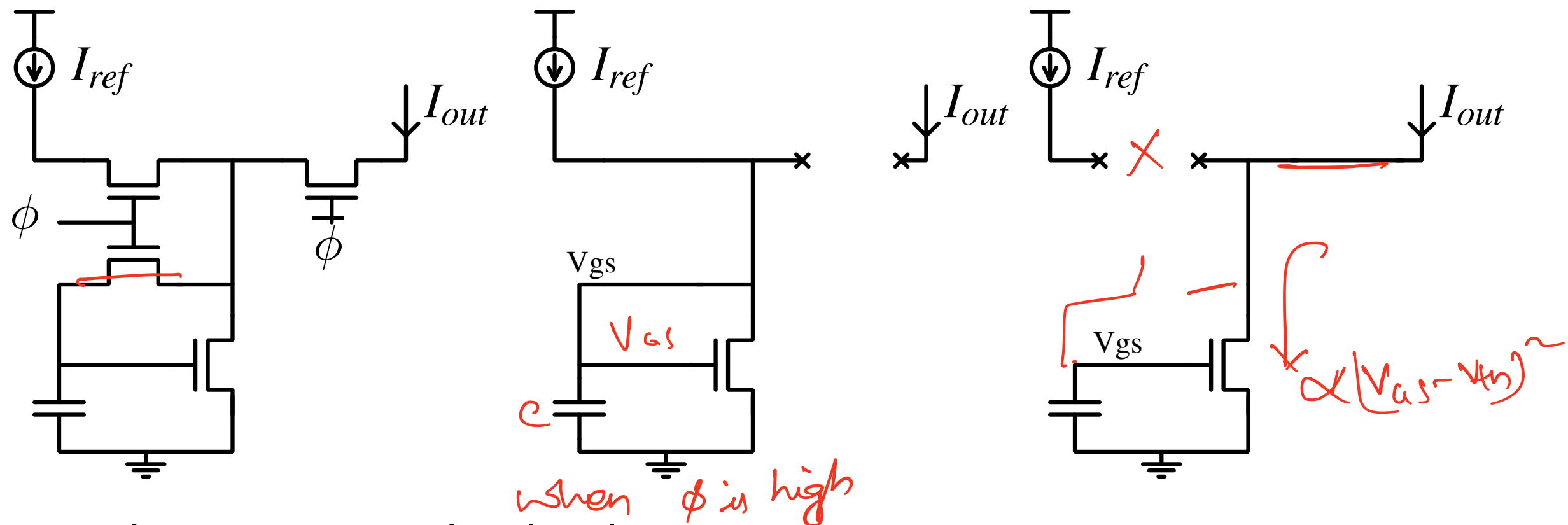
Switched capacitor circuits can be used to improve current mirror matching

- By reusing the same transistor for generating V_{GS} and as the current source transistor, we can get accurate mirroring of current.



Switched capacitor circuits can be used to improve current mirror matching

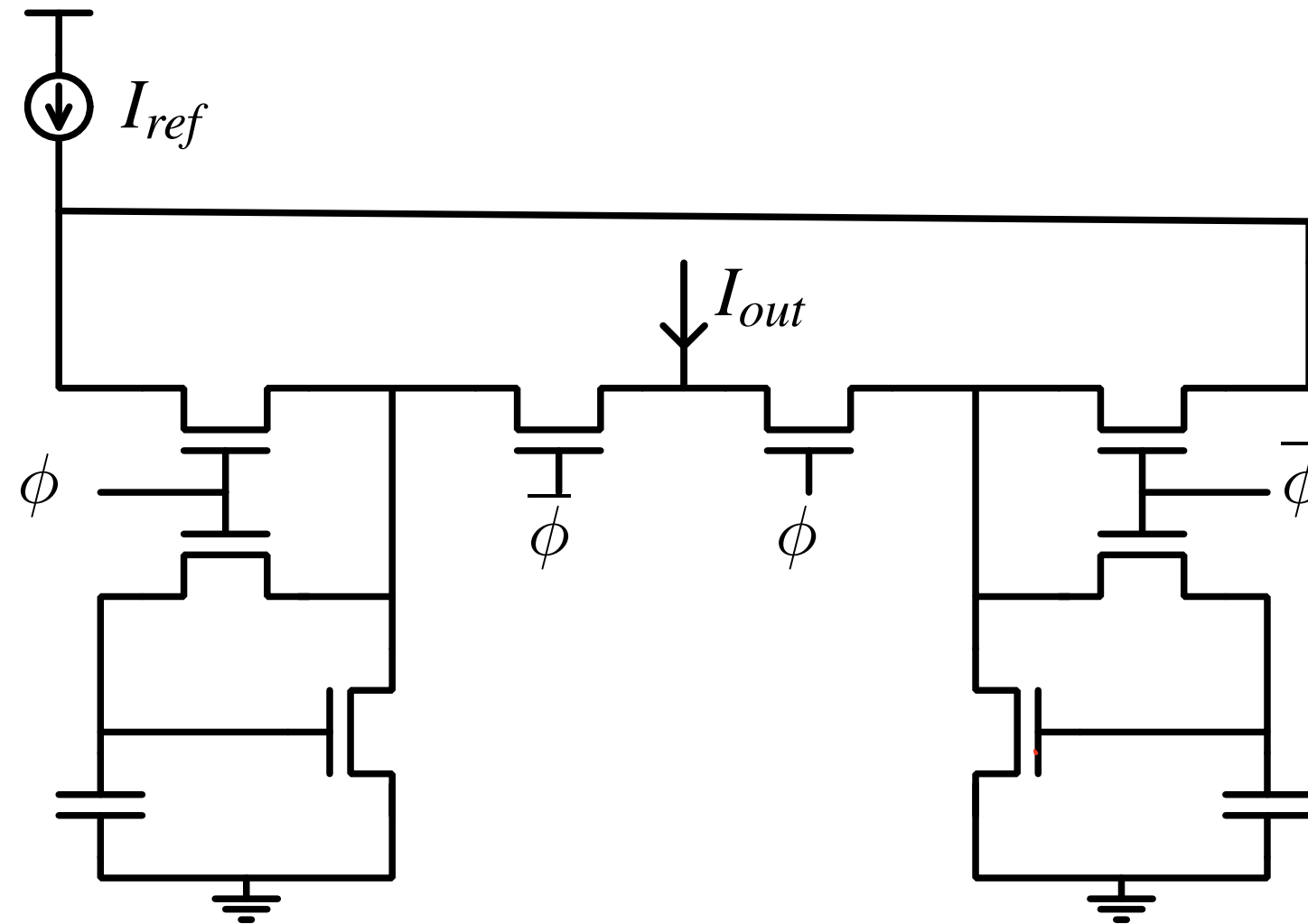
- By reusing the same transistor for generating V_{GS} and as the current source transistor, we can get accurate mirroring of current.



- This current mirror generated pulsed current output.

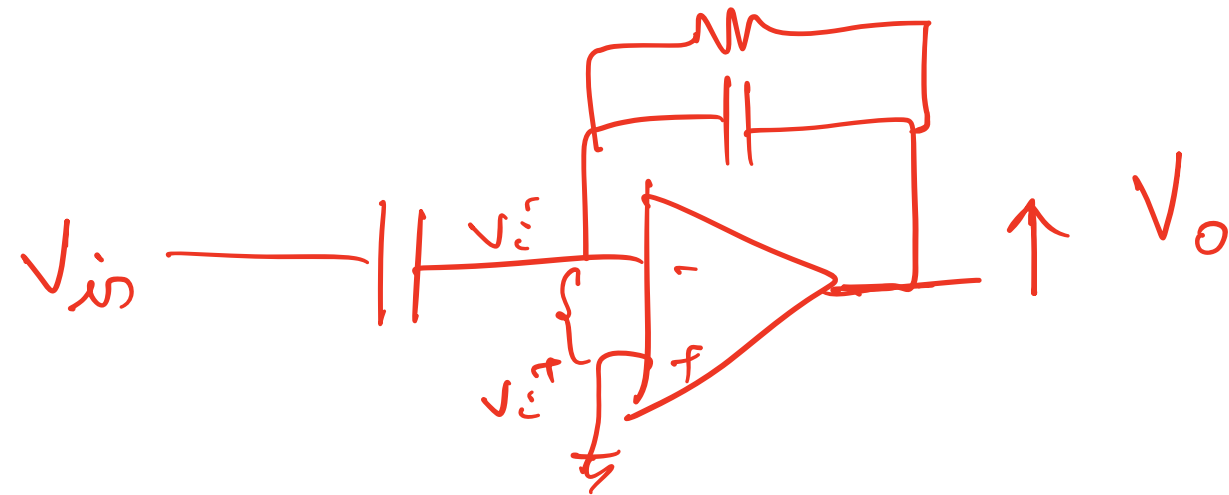
Switched capacitor circuits can be used to improve current mirror matching

- By using two circuits operating in each of the clock phases, continuous operation can be achieved.

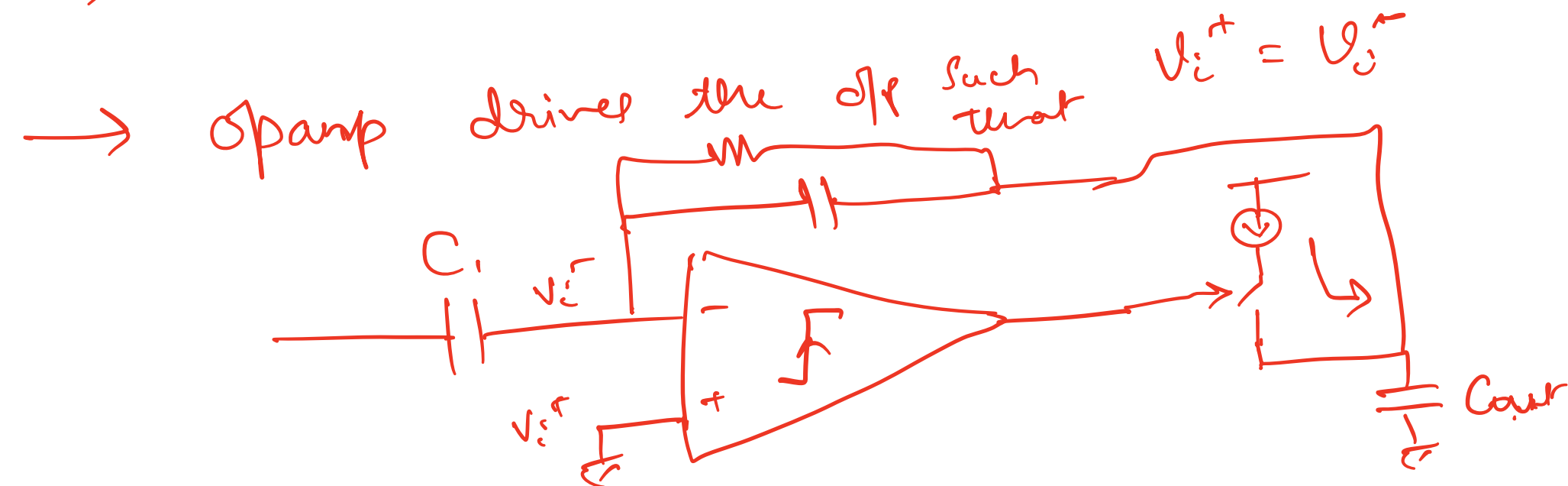


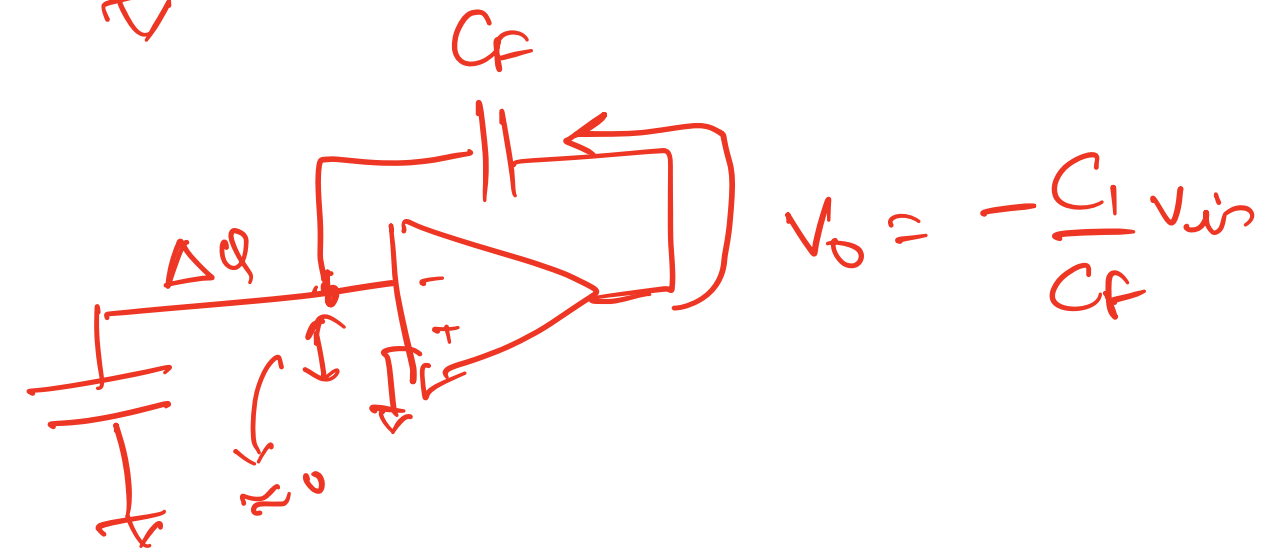
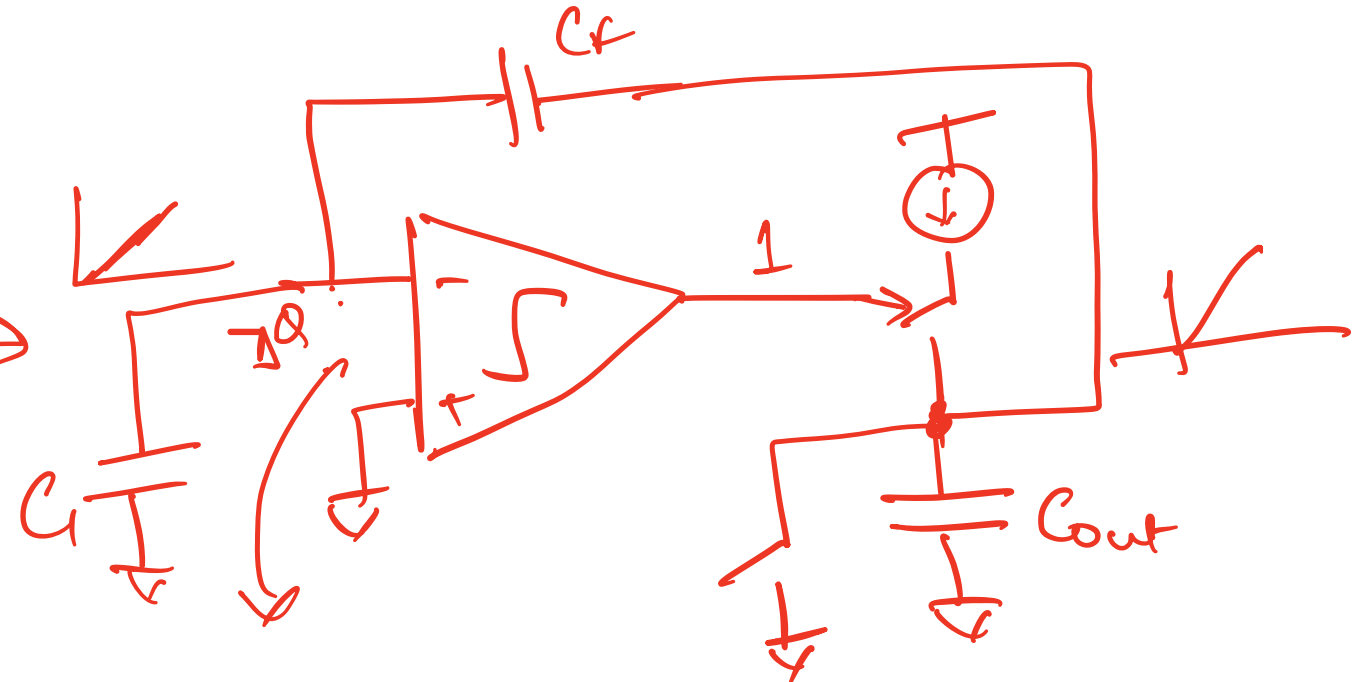
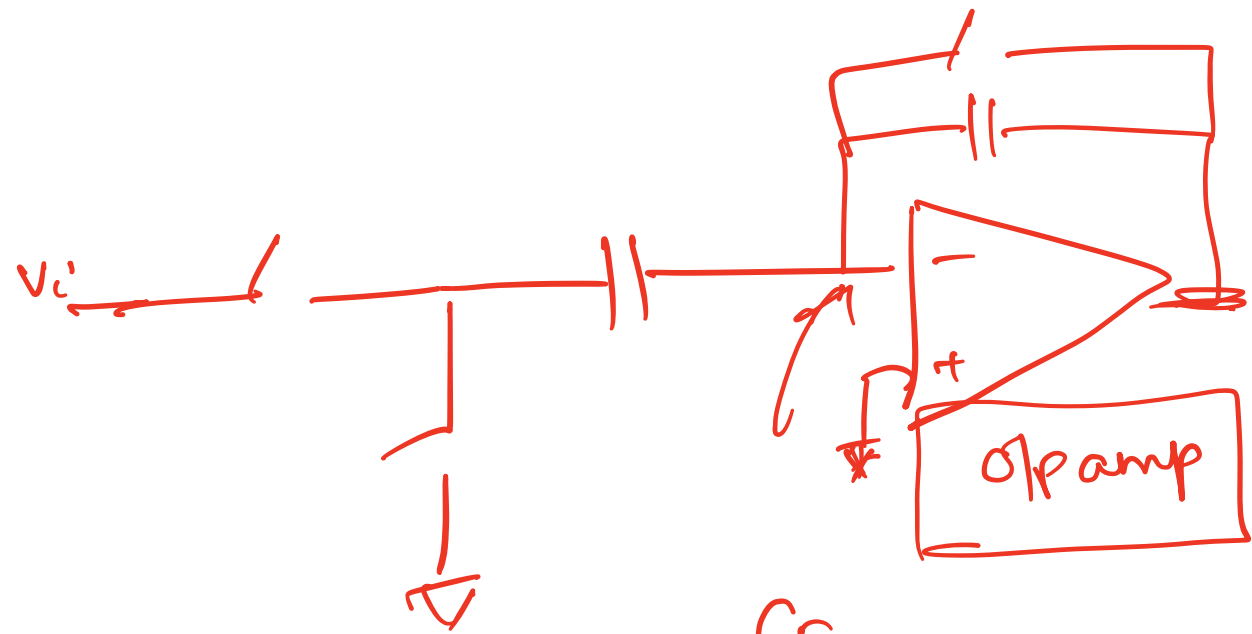
Switch cap circuits eliminate need for resistors in the circuits.

We still need amplifiers.

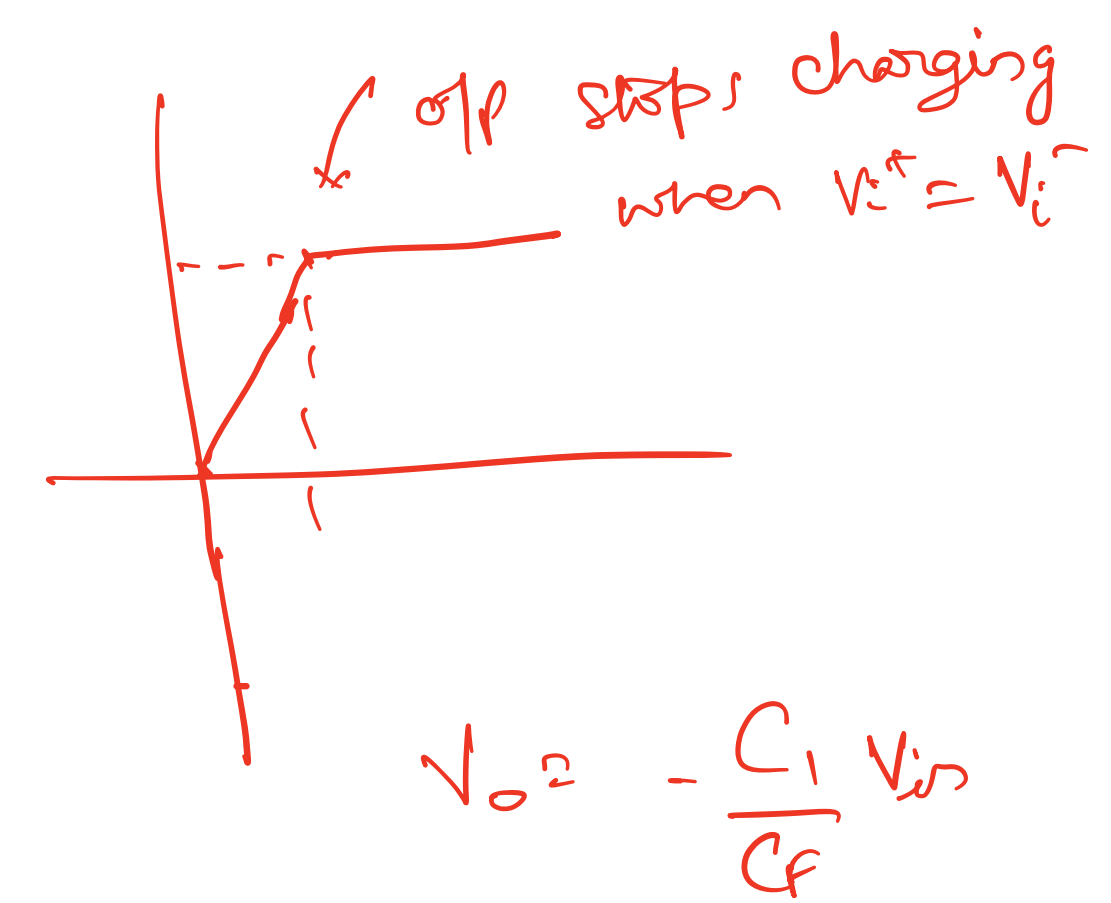
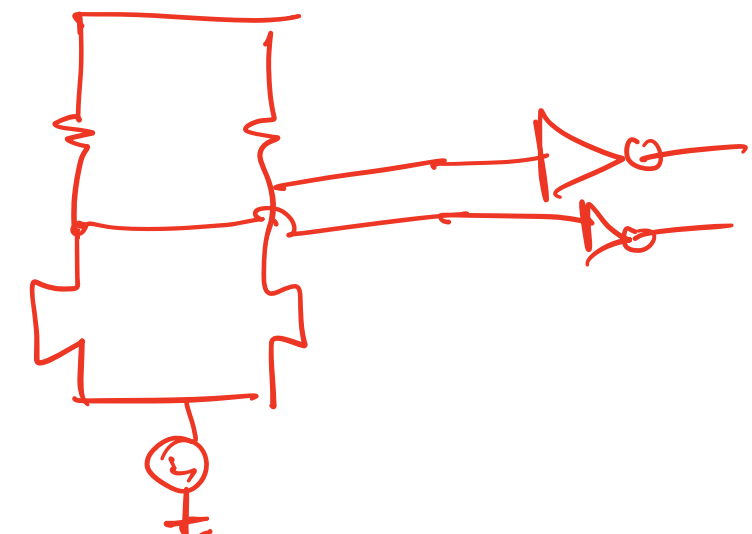


→ opamp integrates error voltage till $V_i^- = V_i^+$

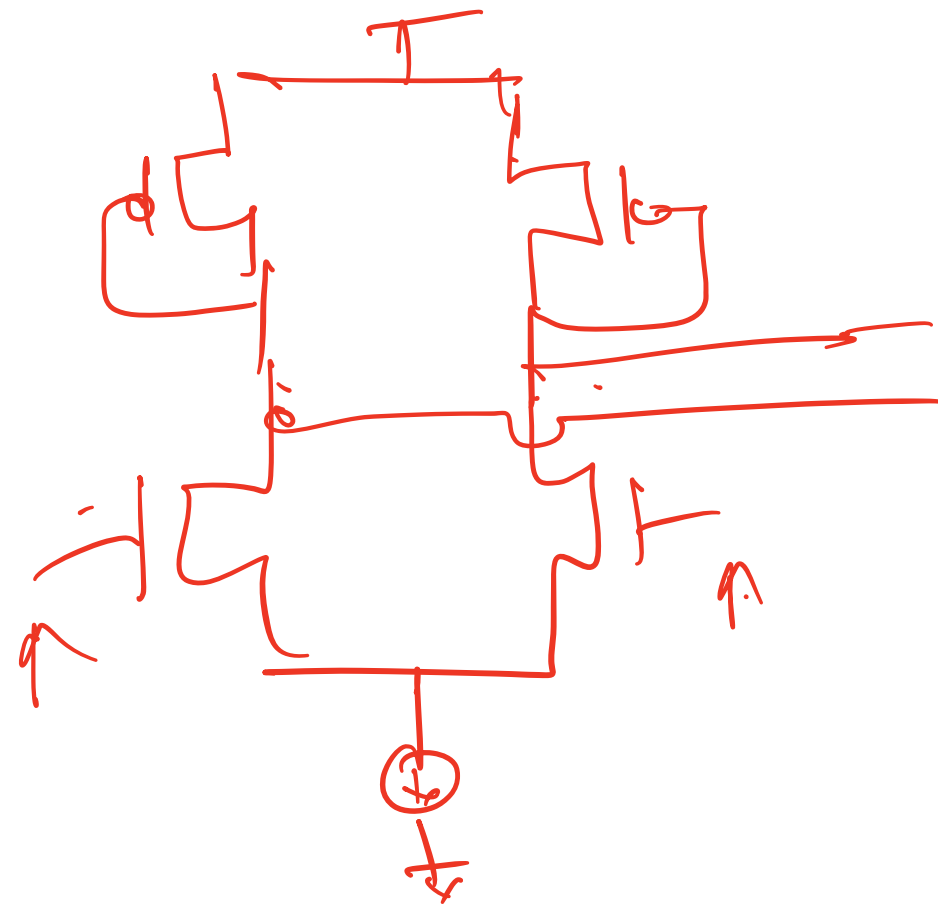




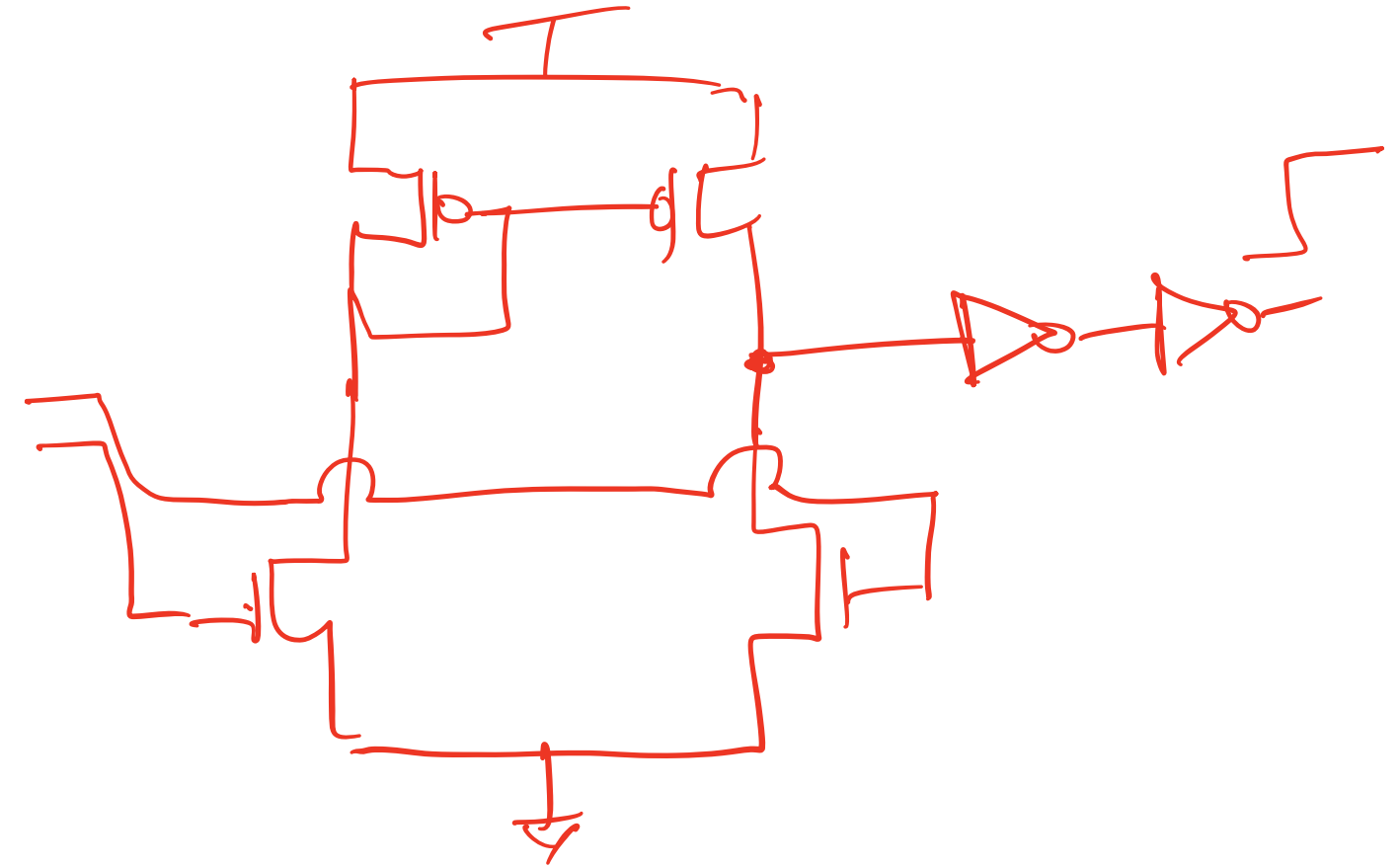
$$V_o = -\frac{C_1}{C_f} V_{in}$$



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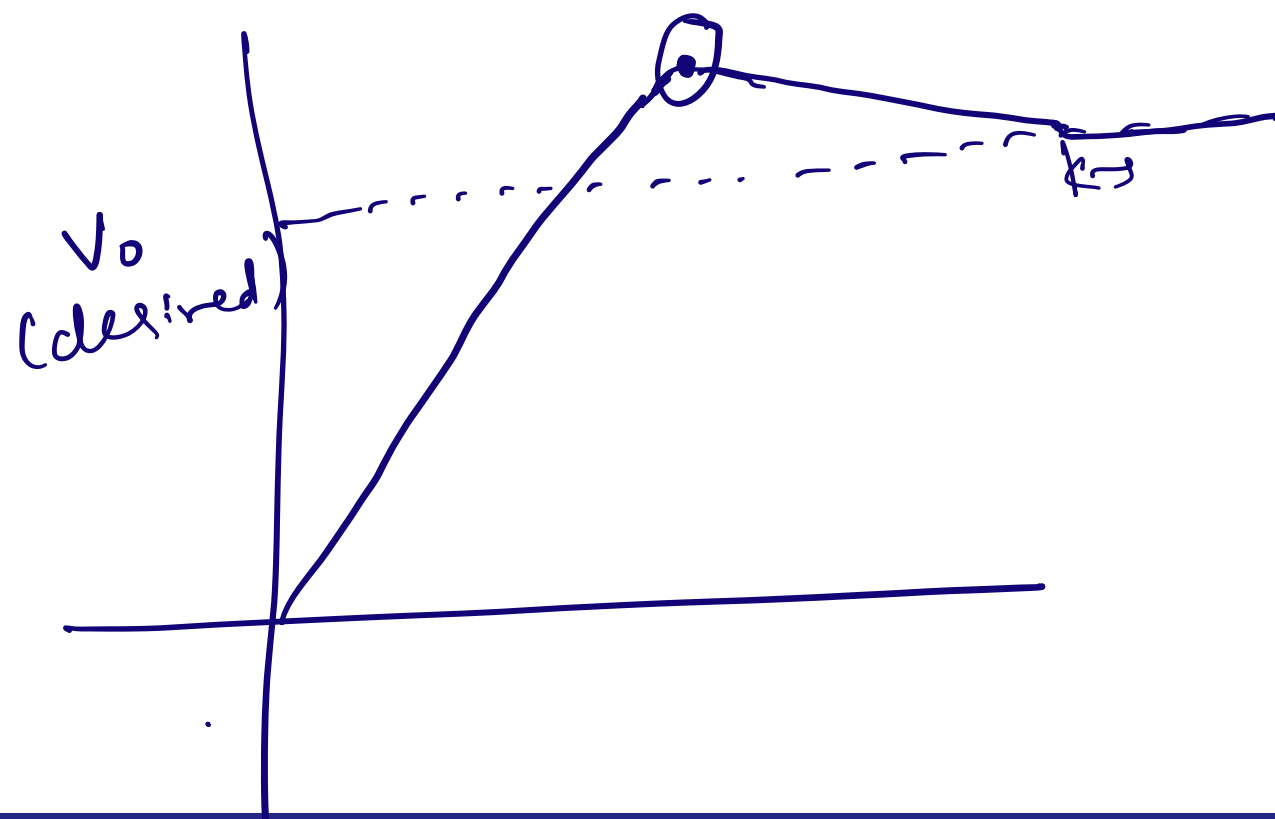
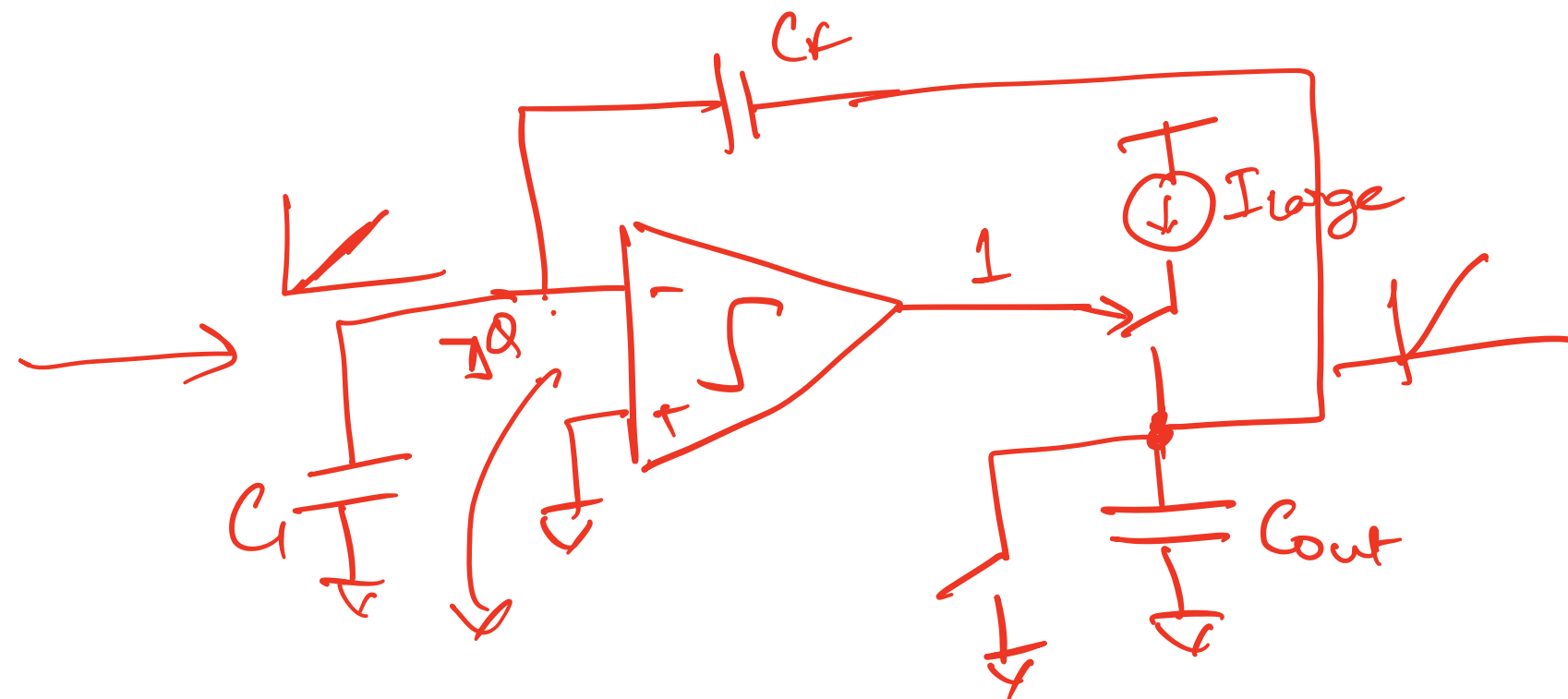
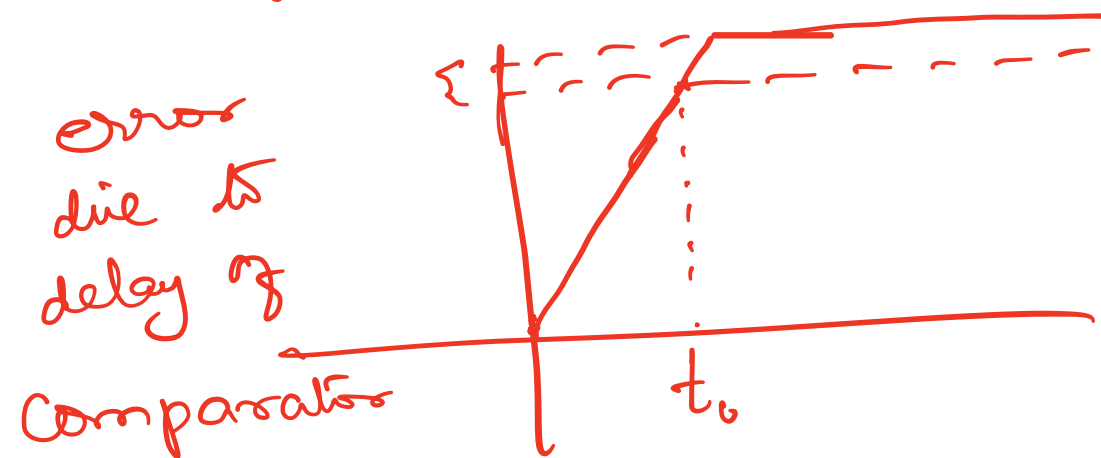


2-3 such
... stages

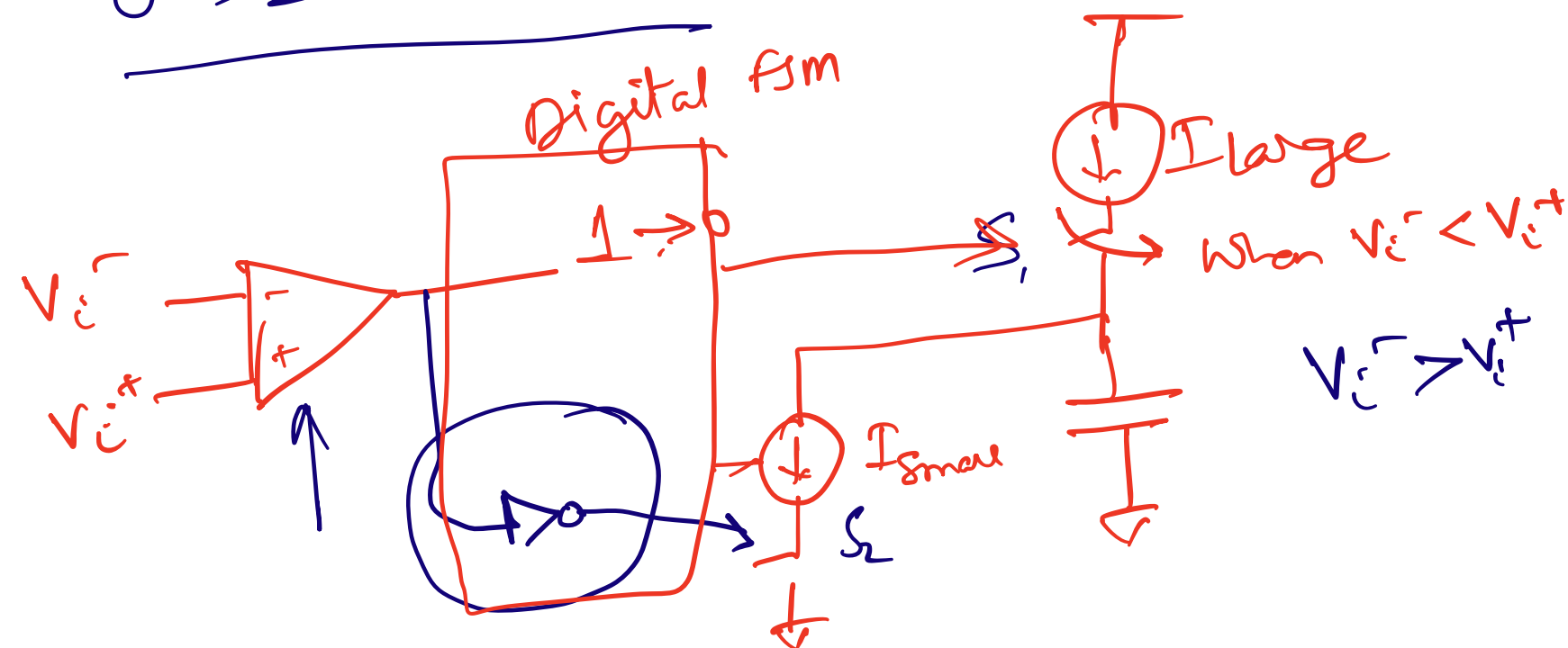


Comparator

Speed?



0 → 1 → 0 → 1



Questions and Discussions

References

- ① Textbook: Behzad Razavi, “Design of Analog CMOS Integrated Circuits”
- ② Textbook: Paul Gray, Paul Hurst, Stephen Lewis, Robert Meyer, “Analysis and Design of Analog Integrated circuits”
- ③ Textbook: R. Jacob Baker, “CMOS circuit design, layout and simulation”

Further reading:

- ① J. K. Fiorenza, T. Sepke, P. Holloway, C. G. Sodini and H. -S. Lee, "Comparator-Based Switched-Capacitor Circuits for Scaled CMOS Technologies," in IEEE Journal of Solid-State Circuits, vol. 41, no. 12, pp. 2658-2668, Dec. 2006, doi: 10.1109/JSSC.2006.884330.